

Kiddies Town ECD & Academy

Technical Handover Pack | Version 2 | 30/08/2026

1. System Overview

A unified full-stack early childhood development (ECD) web platform with real-time portal workspaces for **Parents**, **Teachers** and **Administrators**. Strictly POPIA-compliant data structures (student = parent-consented data), enrolment wizard, attendance registers, progress reports (10 developmental tracks), fee ledger, chat, calendar, weekly-theme curriculum, journal gallery and an admin audit trail.

2. Tech Stack

LAYER	TECHNOLOGY
Frontend	React 19, TypeScript 5.8, Tailwind CSS v4, Vite 6
Routing / State	react-router-dom v7 (lazy, role-protected) · Zustand v5 (auth) · TanStack Query v5 (server data)
Backend	Express 4, esbuild bundling, pino logger, express-rate-limit, helmet + CORS
Database	NeonDB PostgreSQL (serverless) with automatic JSON fallback (<code>data_store.json</code>)
Auth	JWT access + refresh tokens, bcryptjs hashing
PDF / AI	pdfkit (curriculum guide) · optional Gemini key for server-side AI routes

3. Demo Accounts

ROLE	EMAIL	PASSWORD
Parent	parent@kiddiestown.co.za	parent
Teacher	teacher@kiddiestown.co.za	teacher
Admin	admin@kiddiestown.co.za	admin

4. Setup & Running

```
npm install
# optional NeonDB: DATABASE_URL postgresql://user:pass@.../db?sslmode=require
# JWT_SECRET + JWT_REFRESH_SECRET required (min 32 chars)

npm run dev      # dev server (Vite + Express) on port 3000
npm run build    # production bundle -> dist/
npm start        # run the production build (node dist/server.mjs)
```

5. Production Build — Verified

30 Aug 2026: `npm run build` completes; `npm start` boots on :3000; `GET /` serves the SPA; `POST /api/auth/login` (admin/teacher/parent) returns JWT; `GET /api/admin/audit-logs` returns JSON array;

learner attendance POST /api/learners returns 200. Both product fixes below are inside the current dist/build.

6. API Endpoints

AUTH	POST /api/auth/login · /signup · /refresh-token · /change-password
DATA	GET /api/all-data · POST /api/learners · DELETE /api/learners/:id · POST /api/parent-profile · /progress-reports · /payments · /payments/:id/verify · /chats · /events · /themes · /journal · /enrolments · /register
ADMIN	POST /api/admin/create-parent · /send-bulk-emails · /send-arrears-notice · /reset-db GET /api/admin/audit-logs
PDF	GET /api/pdf/guide

7. Database Tables (NeonDB, `kt_` prefix)

users · learners · parent_profile · progress_reports · payments · chats · weekly_themes · school_events · journal_posts · enrolments · registers (JSONB) · audit_logs

Fallback mode (no DATABASE_URL) persists all collections to `data_store.json`.

8. QA Evidence & Defects Fixed (30 Aug 2026)

Automated browser suite: **42/42 PASS** across Public (5), Parent (10), Teacher (11), Admin (15), Cleanup (1).
Evidence: Kiddies-Town-UI-Test-Report.pdf with per-case screenshots.

Fixed

- **Learner schema (attendance 400).** `server/schemas/data.schemas.ts` — null read-backs for `enrolmentApproved/transportNeeded/transportRouteId/transportRouteName` rejected by `.optional()`; now `.nullish()`.
- **Audit-logs endpoint.** `src/components/AdminDashboard.tsx` used `/audit-logs`; corrected to `/admin/audit-logs`.
- **Server crash on public enrolment.** `server/utils/dbHelpers.ts` — operator guard `authReq.user?.email || 'guest'`.

9. Known Issues / Notes

- Rate limiter: 100 req / 15 min per IP on `/api` (see `server/middleware/rateLimiter.ts`).
- Enrolment wizard auto-submits when Step 5 "Continue" is clicked (React node-swap quirk); Step 6 upload/review screen is effectively skipped. Suite tolerates both paths.
- Legacy unused files: `server.ts`, `App.legacy.tsx`. Admin/Parent dashboards are large single components (refactor candidates).
- One benign non-API 404 (missing static asset) observed in console.